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Pre-Lab Questions

1. Compute the transfer function $H(j\omega)$ for the following non-inverting op-amp circuit.

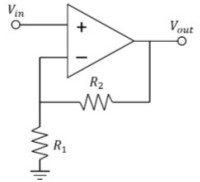


Figure 1: Non-inverting Op-Amp Circuit Diagram

Transfer function

$$H(j\omega) = \frac{V_{out}(j\omega)}{V_{in}(j\omega)}$$

- $Z_{R_1} = R_1$
- $Z_{R_2} = R_2$

$V_{in} = Z_{R_1}$ $\qquad$ $V_{out} = Z_{R_1} + Z_{R_2}$

$$H(j\omega) = \frac{V_{out}(j\omega)}{V_{in}(j\omega)} = \frac{Z_{R_1} + Z_{R_2}}{Z_{R_1}} = \frac{R_1 + R_2}{R_1} \Rightarrow \boxed{1 + \frac{R_2}{R_1}}$$

2. Create the following circuit in PSpice, perform a time-domain simulation and provide a copy of the schematic and the graphed results in your prelab 4 submission.

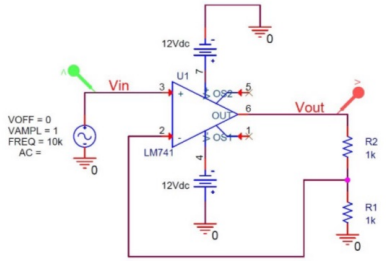
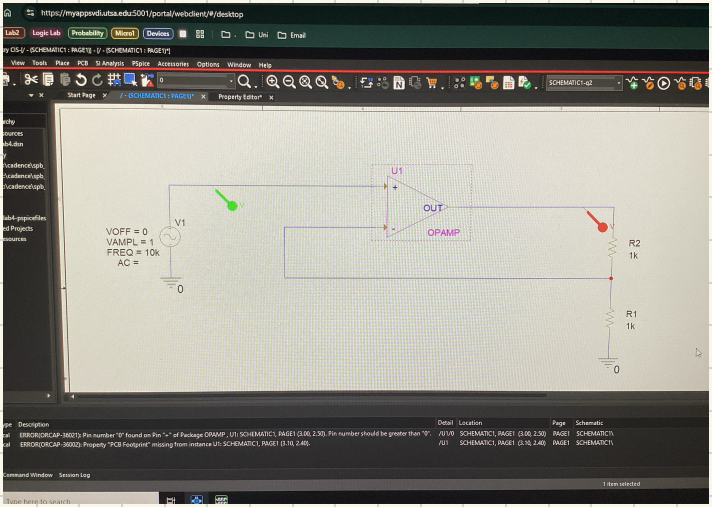
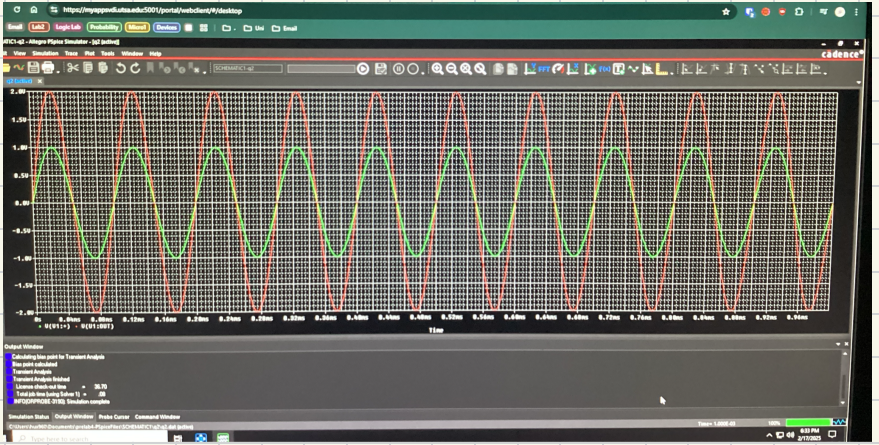


Figure 2: Non-inverting Op-amp Schematic with Voltage Probes

Note that:

- No connects need to be placed on pins 1 and 5 of the op-amp.
- Voltage sources need to be oriented such that negative voltage is supplied to pin 4 and positive voltage is supplied to pin 7.
- Under *Simulation Profile Settings* use 1ms for the 'Run To Time' setting, and use 1us for the 'Maximum Step Size' setting.


U1
OUT
OPAMP
V1
VOFF = 0
VAMPL = 1
FREQ = 10k
AC =
R2
1k
R1
1k
0
0



3. Modify the circuit from the previous prelab question by changing the voltage source parameters as well as the simulation profile setting and perform a frequency-domain simulation. Provide a copy of the schematic and the graphed results in your prelab 4 submission.

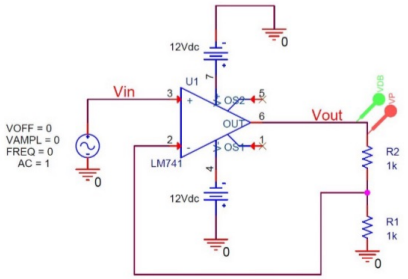
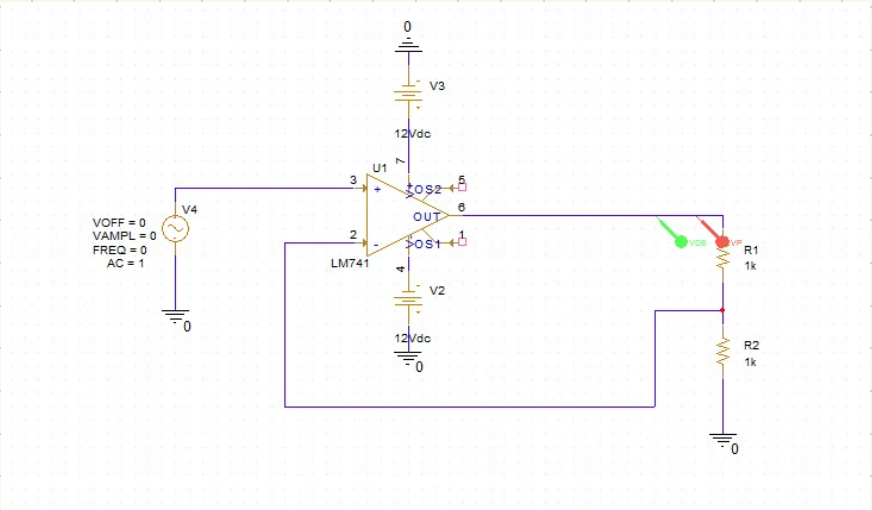
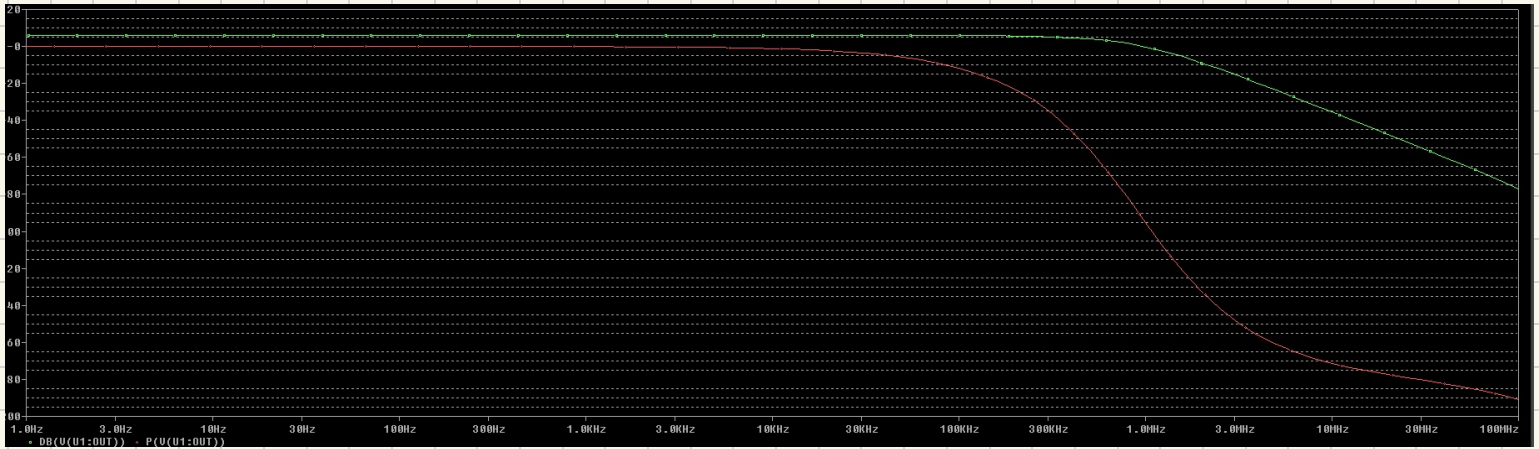


Figure 3: Non-inverting Op-amp Schematic with Gain and Phase Probes

- Make sure to change the simulation profile setting to AC Sweep/Noise and use 1 as the 'Start Frequency' parameter, use "100mega" as the 'End Frequency' parameter, and use 10 as the 'Points/Decade' parameter.
- Use the 'dB Magnitude of Voltage' and 'Phase of Voltage' probes from the **PSpice > Markers > Advanced** list.

[Provide the Pre-Lab answers separate from the lab report, and show your work.]


0
V3
12Vdc
U1
7
3
OS2
5
V4
VOFF = 0
VAMPL = 0
FREQ = 0
AC = 1
OUT
6
2
OS1
1
LM741
4
V2
12Vdc
0
VDB
VP
R1
1k
R2
1k
0


20
0
20
40
60
80
00
20
40
60
80
00
1.0Hz
3.0Hz
10Hz
30Hz
100Hz
300Hz
1.0KHz
3.0KHz
10KHz
30KHz
100KHz
300KHz
1.0MHz
3.0MHz
10MHz
30MHz
100MHz
DB(U(U1:OUT))
P(U(U1:OUT))